

Apollo Lunar Module — One-Shot Motion-Planning OCP

Optimal Control Problem Formulation

State vector (17)

$$\mathbf{x} = \left[\underbrace{x_E, y_E, z_E}_{\text{position } \mathbf{r}_E}, \underbrace{u, v, w}_{\text{body vel. } \mathbf{v}_b}, \underbrace{\phi, \theta, \psi}_{\text{Euler } \boldsymbol{\Theta}}, \underbrace{p, q, r}_{\text{body rates } \boldsymbol{\omega}}, \underbrace{T, \delta_p, \dot{\delta}_p, \delta_y, \dot{\delta}_y}_{\text{actuator states}} \right]^\top$$

Frames: E is the (flat-Moon) surface frame, B the body frame (x fwd, y right, z down); $z_E \leq 0$ is above ground.

Input vector (19)

$$\mathbf{u} = \left[T_c, \delta_{p,c}, \delta_{y,c}, f_1, f_2, \dots, f_{16} \right]^\top$$

where $T_c, \delta_{p,c}, \delta_{y,c}$ are the commanded thrust and pitch/yaw gimbal angles, and $f_1, \dots, f_{16}$ are the individual RCS thruster forces.

State equations $\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u})$

With the shorthand $c_* = \cos *$, $s_* = \sin *$, $t_* = \tan *$, and $m, J = \text{diag}(I_{xx}, I_{yy}, I_{zz})$, g the mass, inertia and lunar gravity, the continuous dynamics are

$$\dot{\mathbf{r}}_E = C_{E/B}(\boldsymbol{\Theta}) \mathbf{v}_b \quad (\text{navigation}) \quad (1)$$

$$\dot{\mathbf{v}}_b = -\boldsymbol{\omega} \times \mathbf{v}_b + \mathbf{g}_b + \frac{1}{m}(\mathbf{F}_T + \mathbf{F}_{\text{rcs}}) \quad (\text{translation, body frame}) \quad (2)$$

$$\dot{\boldsymbol{\Theta}} = E(\boldsymbol{\Theta}) \boldsymbol{\omega} \quad (\text{attitude kinematics}) \quad (3)$$

$$\dot{\boldsymbol{\omega}} = J^{-1}(\mathbf{M} - \boldsymbol{\omega} \times J \boldsymbol{\omega}) \quad (\text{Euler's equations}) \quad (4)$$

closed by the actuator states

$$\dot{T} = \frac{T_c - T}{\tau_T}, \quad \ddot{\delta}_p = \omega_n^2(\delta_{p,c} - \delta_p) - 2\zeta\omega_n\dot{\delta}_p, \quad \ddot{\delta}_y = \omega_n^2(\delta_{y,c} - \delta_y) - 2\zeta\omega_n\dot{\delta}_y.$$

Rotation matrix (body $\rightarrow$ surface, 3–2–1 Euler):

$$C_{E/B} = \begin{bmatrix} c_\theta c_\psi & s_\phi s_\theta c_\psi - c_\phi s_\psi & c_\phi s_\theta c_\psi + s_\phi s_\psi \\ c_\theta s_\psi & s_\phi s_\theta s_\psi + c_\phi c_\psi & c_\phi s_\theta s_\psi - s_\phi c_\psi \\ -s_\theta & s_\phi c_\theta & c_\phi c_\theta \end{bmatrix}, \quad E = \begin{bmatrix} 1 & s_\phi t_\theta & c_\phi t_\theta \\ 0 & c_\phi & -s_\phi \\ 0 & s_\phi/c_\theta & c_\phi/c_\theta \end{bmatrix}.$$

Forces and moments. The gimballed engine thrust, gravity, and the coupled Coriolis term expand (component-wise) as

$$\mathbf{F}_T = \begin{bmatrix} T s_{\delta_p} \\ -T c_{\delta_p} s_{\delta_y} \\ -T c_{\delta_p} c_{\delta_y} \end{bmatrix}, \quad \mathbf{g}_b = g \begin{bmatrix} -s_\theta \\ s_\phi c_\theta \\ c_\phi c_\theta \end{bmatrix}, \quad \boldsymbol{\omega} \times \mathbf{v}_b = \begin{bmatrix} qw - rv \\ ru - pw \\ pv - qu \end{bmatrix}, \quad \boldsymbol{\omega} \times J \boldsymbol{\omega} = \begin{bmatrix} (I_{zz} - I_{yy})qr \\ (I_{xx} - I_{zz})rp \\ (I_{yy} - I_{xx})pq \end{bmatrix}.$$

The RCS force $\mathbf{F}_{\text{rCS}}$ and total moment $\mathbf{M}$ are

$$\begin{bmatrix} \mathbf{F}_{\text{rCS}} \\ \mathbf{M}_{\text{rCS}} \end{bmatrix} = B \mathbf{f}, \quad \mathbf{M} = \underbrace{\begin{bmatrix} -d_z T_y \\ d_z T_x \\ 0 \end{bmatrix}}_{\text{gimbal offset } d_z=2.5 \text{ m}} + \mathbf{M}_{\text{rCS}},$$

where T_x, T_y are the first two components of $\mathbf{F}_T$, and B is the fixed 6×16 RCS allocation matrix

$$B = [\mathbf{d}_i; \mathbf{r}_i \times \mathbf{d}_i]_{i=1}^{16}, \quad \mathbf{r}_i = \text{thruster position}, \quad \mathbf{d}_i = \text{unit fire direction}.$$

Discretisation

Direct multiple shooting on a uniform grid of $N = 80$ intervals, $\Delta t = 1 \text{ s}$ (horizon 80 s), with a 4th-order Runge–Kutta integrator Φ_{RK4} :

$$\mathbf{x}_0 = \mathbf{x}_{\text{init}}, \quad \mathbf{x}_{k+1} = \Phi_{\text{RK4}}(\mathbf{x}_k, \mathbf{u}_k; \Delta t), \quad k = 0, \dots, N-1.$$

Cost function

$$J = \sum_{k=0}^{N-1} \left[(\mathbf{x}_k - \mathbf{x}_{\text{ref}})^\top Q_s (\mathbf{x}_k - \mathbf{x}_{\text{ref}}) + (\mathbf{u}_k - \mathbf{u}_{\text{ref}})^\top R (\mathbf{u}_k - \mathbf{u}_{\text{ref}}) \right] \\ + \sum_{k=1}^{N-1} \Delta \mathbf{u}_k^\top R_\Delta \Delta \mathbf{u}_k + (\mathbf{x}_N - \mathbf{x}_{\text{ref}})^\top Q_f (\mathbf{x}_N - \mathbf{x}_{\text{ref}})$$

with $\Delta \mathbf{u}_k = \mathbf{u}_k - \mathbf{u}_{k-1}$. The middle sum is a *control-rate* (move-suppression) penalty: unlike R , which penalises command *magnitude*, R_Δ penalises how fast the commands *change*, damping gimbal-command oscillation. The references are $\mathbf{x}_{\text{ref}} = \mathbf{0}$ except the terminal altitude (see cut-off, below), and

$$\mathbf{u}_{\text{ref}} = [T_{\text{hover}}, 0, 0, 0, \dots, 0]^\top, \quad T_{\text{hover}} = m g.$$

Diagonal weights (element-wise; actuator states carry zero weight), grouped as [pos | vel | att | rate | act]:

$$Q_s = \text{diag}(20, 20, 30 \mid 60, 60, 60 \mid 20, 20, 1 \mid 30, 30, 30 \mid 0, 0, 0, 0, 0) \\ Q_f = \text{diag}(5000, 5000, 8000 \mid 4000, 4000, 4000 \mid 6000, 6000, 400 \mid 10^4, 10^4, 10^4 \mid 0, 0, 0, 0, 0) \\ R = \text{diag}(10^{-7}; 8, 8; \underbrace{5 \cdot 10^{-4}, \dots}_{16}), \quad R_\Delta = \text{diag}(10^{-6}; 4000, 4000; \underbrace{10^{-3}, \dots}_{16})$$

Path constraints

$$\text{state } (\forall k = 0, \dots, N) : \quad \begin{aligned} z_E &\leq 0, & -V_{\text{max}} &\leq u, v, w \leq V_{\text{max}}, & V_{\text{max}} &= 60 \text{ m/s} \\ -\Theta_{\text{max}} &\leq \phi, \theta, \psi \leq \Theta_{\text{max}}, & \Theta_{\text{max}} &= 45^\circ, \\ -\Omega_{\text{max}} &\leq p, q, r \leq \Omega_{\text{max}}, & \Omega_{\text{max}} &= 10^\circ/\text{s}, \\ T_{\text{min}} &\leq T \leq T_{\text{max}}, & -\delta_{\text{max}} &\leq \delta_p, \delta_y \leq \delta_{\text{max}}, & \delta_{\text{max}} &= 6^\circ \end{aligned}$$

$$\text{input } (\forall k = 0, \dots, N-1) : \quad \begin{aligned} T_{\text{min}} &\leq T_c \leq T_{\text{max}}, & -\delta_{\text{max}} &\leq \delta_{p,c}, \delta_{y,c} \leq \delta_{\text{max}}, \\ 0 &\leq f_i \leq F_{\text{rCS}}, \quad i = 1, \dots, 16, & F_{\text{rCS}} &= 445 \text{ N}, \end{aligned}$$

with $[T_{\text{min}}, T_{\text{max}}] = [4560, 45040] \text{ N}$.

Two-phase landing with engine cut-off

The powered OCP above targets a low *contact* altitude $z_E = -h_{\text{contact}}$ ($h_{\text{contact}} = 1$ m) over the pad, rather than the ground. At contact the DPS is cut and the last stretch to the surface is an *analytic*, engine-off ballistic settle: from the powered terminal state the thrust is hard-cut ($T := 0$, gimbals zeroed, RCS off) and $\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{0})$ is integrated with fine steps until $z_E = 0$. For a near-rest contact the touchdown speed is

$$v_{\text{td}} = \sqrt{w_c^2 + 2g h_{\text{contact}}},$$

w_c being the vertical descent rate at cut-off. This mirrors the real DPS shutdown at contact and keeps the OCP well-posed near $z = 0$ (no thrust gating or hover pathology).

Remarks.

- Thrust and gimbal limits appear *twice* — as state bounds on the actual T, δ_p, δ_y and as input bounds on the commands — because the actuators are dynamic states driven by their commands.
- The attainable RCS wrench set is the image of the per-thruster box $0 \leq f_i \leq F_{\text{rps}}$ under B (a coupled zonotope), not an axis-aligned box.
- There is *no* terminal equality constraint; the final state is driven only softly by Q_f toward the contact condition.